Project ParkOrbit

Peer-to-Peer Urban Parking Marketplace & Geospatial Mobility Platform

DESIGN THINKING REPORT

Phase 1: Empathize

1. Goal

To understand the real-life experiences of daily urban commuters, electric vehicle (EV) drivers, and residential or commercial property owners when searching for or sharing parking spaces across high-density areas in Pune, and to identify the physical, informational, trust, and navigational barriers they encounter.

2. Objectives

1. **Understand user needs:** Learn about the daily parking and charging requirements of two-wheeler and four-wheeler commuters, EV owners, and homeowners or housing societies with idle parking bays.
2. **Identify pain points:** Discover barriers such as on-street cruising congestion, lack of real-time slot availability, unclear height and dimensional clearances, cash haggling, and security concerns around letting strangers into private gates.
3. **Study existing practices:** Understand how drivers currently hunt for curbside parking in busy corridors (Koregaon Park, Baner, FC Road, Hinjawadi) and how property owners manage vacant driveways or stilt bays.
4. **Understand user expectations:** Identify what verified information drivers need before travelling (hourly pricing, covered status, EV charger wattage, gate access instructions) and what controls hosts need to list spots safely.
5. **Gather user insights:** Collect field observations and user feedback to guide the human-centred design of the ParkOrbit solution.

3. Target Users

1. Daily urban commuters and vehicle drivers (hatchbacks, sedans, SUVs, and two-wheelers)
2. Electric Vehicle (EV) drivers seeking guaranteed destination charging and covered bays
3. Homeowners, gated villa residents, and commercial property managers (Hosts) with underutilized parking spaces
4. Community and platform Trust & Safety moderators managing verification and access disputes

4. Activities to Achieve the Objectives

Activity	What we will do
User interviews	Ask daily commuters, EV owners, and Pune property owners about their experiences finding parking and managing vacant driveways.
Observation	Observe peak-hour curb cruising, parking signage, gate entry bottlenecks, and daytime empty residential bays across busy Pune hubs.
Empathy mapping	Organize findings into what drivers and property hosts say, think, do, and feel during daily parking and hosting scenarios.
Pain-point identification	Identify recurring parking search, vehicle dimension, pricing, and gate security challenges from the collected evidence.
Requirement gathering	Determine what geospatial, pricing, and access verification information users need to reserve and host parking spaces confidently.

5. Interview Questions

1. What difficulties do you face when trying to find parking in busy Pune localities like Koregaon Park, Baner, FC Road, or Hinjawadi?
2. How do you currently check whether safe parking or an EV charger is available before driving to an unfamiliar area?
3. Have you encountered blocked driveways, low-clearance basements unsuitable for SUVs, unverified parking rates, or cash-only attendants?
4. What specific information (live hourly rates, bay dimensions, CCTV coverage, or gate entry instructions) would help you plan and book parking before leaving home?
5. If you own an empty residential driveway or society bay during office hours, what security or management concerns prevent you from renting it out?
6. What would make checking into a private parking spot and reporting access or overstay problems easier and more trustworthy?

6. Expected Outcome

- A better understanding of both drivers' parking needs and hosts' security expectations.
- A structured record of commuter and homeowner interview responses and field observations.
- An empathy map representing driver frustration while circling blocks and host hesitation around gate security.
- A prioritized list of the most significant urban parking discovery, booking, and trust barriers.
- Initial insights to help clearly define the core problem in Phase 2.

Phase 2: Define

1. Goal

To clearly define the main urban parking discovery and idle space utilization problem faced by commuters and property owners in high-density neighbourhoods, based on user needs and evidence gathered during the Empathize phase.

2. Objectives

1. **Analyze user research:** Review interview responses and field observations from Phase 1 to identify recurring difficulties faced by drivers and hosts.
2. **Identify the primary pain point:** Determine which informational and trust barrier creates the greatest friction in connecting searching drivers with empty private parking bays.
3. **Identify user needs:** Establish what real-time availability, pricing, vehicle compatibility, and gate security features users need to book and host spaces independently.
4. **Formulate a problem statement:** Express the identified problem clearly, specifying the target users, their needs, and the underlying challenge.
5. **Develop a How Might We question:** Frame the problem as an open-ended question that guides creative solutions in Phase 3, Ideate.

3. Identify the Core Problem

Potential problems to investigate include:

- Lack of reliable, real-time geospatial information about available parking bays and hourly prices near busy destinations.
- High daytime idle capacity (45–60%) in private residential driveways, bungalows, and society stilt spaces while drivers circle blocks outside.
- Uncertainty about bay dimensions, roof coverage, SUV height clearance, and working EV charging points.
- Security concerns and gate friction when sharing access codes or handling cash with unfamiliar drivers.
- Absence of a structured reporting and moderation mechanism to resolve blocked bays, inaccurate listings, or overstays.

4. Define the Problem Statement

Our proposed problem statement is:

> **Urban drivers and EV commuters in dense Pune corridors find it difficult to locate safe, predictable, and affordable parking because reliable, real-time information about available spaces, vehicle clearances, and EV chargers is not readily accessible—even though thousands of private driveways and society bays sit empty nearby. Both drivers and property owners need a trusted, accessible way to discover, book, verify gate access, and manage private parking spaces by the hour.**

5. Develop the "How Might We" Question

> *"How might we help urban commuters find reliable, real-time information about available private parking spaces and enable homeowners to share their idle driveways securely so that drivers can park with greater confidence and hosts can earn passive income effortlessly?"*

Phase 3: Ideate

1. Goal

To generate, explore, and evaluate innovative, practical solutions that help urban commuters find reliable parking and EV charging spaces while enabling property owners to safely list and manage their idle parking bays.

2. Objectives

1. **Brainstorm solutions:** Generate multiple ideas to address the defined urban parking and space-sharing problem without judging them prematurely.

2. **Explore alternatives:** Consider physical hardware, community-based directories, and digital marketplace solutions.
3. **Encourage creativity:** Combine and improve ideas to identify frictionless, high-trust approaches.
4. **Evaluate feasibility:** Compare solutions based on usability, cost, accessibility, and potential urban and social impact.
5. **Select the best idea:** Choose a promising solution for development in Phase 4, Prototype.

3. Brainstorm Possible Solutions

Idea 1: Live Parking Availability Map
An interactive digital map showing available parking spots, hourly rates, covered status, and EV charging points across Pune neighbourhoods.

Idea 2: Community Slot & Reporting Directory
A neighbourhood noticeboard and reporting tool where residents post vacant slot timings and commuters flag blocked entrances or poor signage.

Idea 3: Smart IoT Bollard & RFID Hardware
Physical automated parking locks and RFID scanners installed in private driveways that unlock when a registered commuter vehicle arrives.

Idea 4: ParkOrbit — P2P Map, Booking & Gate PIN App
A unified web/mobile marketplace combining live ₹/hr map pins, instant hourly booking, a 4-digit Gate PIN pass, host pin-drop onboarding, and admin moderation.

4. Comparing the Ideas

Idea	Usability	Cost	Social Impact
Live Parking Availability Map	High potential	Low–medium	Medium–high
Community Slot & Reporting Directory	Medium	Low	Medium
Smart IoT Driveway Bollard Hardware	Medium	High	Medium–high
ParkOrbit — Combined P2P Map, Booking & Gate PIN Application	**High potential**	**Low–medium**	**High**

5. Select the Best Solution

After evaluating the possible solutions based on usability, affordability, accessibility, and urban impact, we selected **ParkOrbit — Peer-to-Peer Map, Booking & Gate PIN Application** as the best solution.

Selected Solution: ParkOrbit Web & Mobile Application — A peer-to-peer urban parking marketplace designed to help commuters find and reserve verified private parking spots by the hour and empower homeowners to monetize empty bays with secure PIN check-ins.

Key Features:

1. **Geospatial Discovery Map:** Displays available private spaces across Pune neighbourhoods (Koregaon Park, Baner, Hinjawadi, Kothrud, Viman Nagar, FC Road, Wakad, Magarpatta) with live ₹/hr pins and vehicle/space filters.
2. **Space Specifications & Instant Booking:** Shows verified host badges, exact bay dimensions (length, width, height clearance), amenities (24/7 CCTV, 7.4kW EV charger), and transparent fee calculation with collision checks.
3. **Digital Parking Pass & Gate PIN Check-In:** Issues a one-time 4-digit Gate PIN (e.g., 4829) and gate entry instructions after reservation confirmation, protecting host privacy and eliminating cash haggling.
4. **Host Command Center & Pin-Drop Onboarding:** Enables homeowners to list a spot in under three minutes by dragging a map pin, configuring hourly rates, and tracking gross bookings, 12% platform fee,

and net payouts.

5. **Trust, Safety & Barrier Reporting Moderation:** Allows users to report access issues and gives admins a console to suspend listings, resolve disputes, or update marketplace fees.

Why We Selected This Solution:

- **Usability & Affordability:** Unifies discovery, booking, and 4-digit Gate PIN verification in software without expensive IoT hardware.
- **Accessibility & Practicality:** Helps drivers lock in dimension-matched parking and EV charging before starting their trip via a responsive Next.js & Leaflet web app.
- **Social Impact:** Reduces on-street cruising congestion, cuts vehicle emissions, supports EV adoption, and generates passive household income.

Phase 4: Prototype

1. Goal

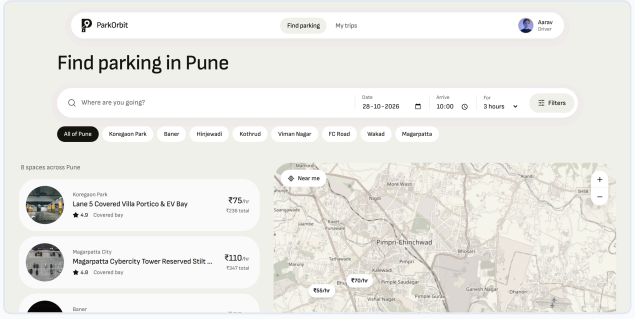
To create a functional prototype and visual representation of **ParkOrbit — Peer-to-Peer Urban Parking Marketplace** to demonstrate how drivers discover and book private parking spots, verify access via Gate PINs, and how hosts and admins manage spaces.

2. Objectives

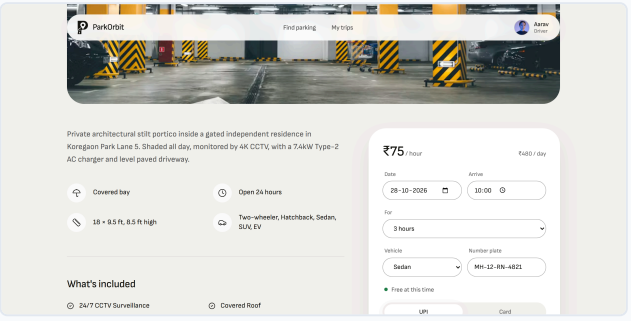
- Develop interactive screens and clean visual views of the main workflows of the ParkOrbit application.
- Demonstrate how drivers search across Pune localities, filter by vehicle type, and inspect live ₹/hr pins on an interactive map.
- Show how users inspect dimensional clearances, EV charger availability, and transparent fee breakdowns before booking.
- Illustrate the Digital Parking Pass with a 4-digit Gate PIN and the Host Command Center for managing listings and earnings.

3. Prototype Development

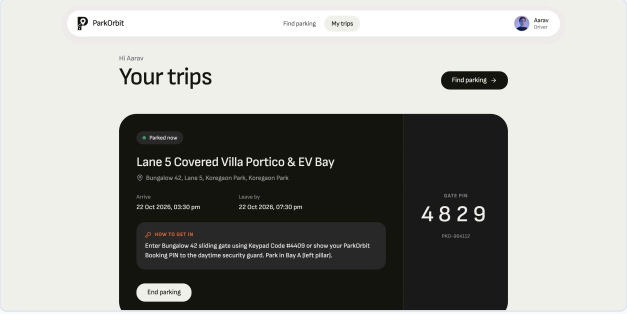
A functional full-stack prototype of ParkOrbit was developed using Next.js 15, React 19, Tailwind CSS, and Leaflet with Carto Voyager map tiles. The interface uses a high-contrast warm alabaster and signal-orange palette with clear iconography and a 1-click multi-persona switcher (Driver, Host, Admin).



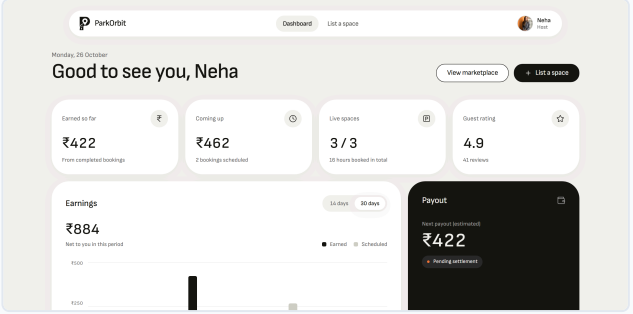
Screen 1: Geospatial Discovery Map (/find) with Pune locality filters and live ₹/hr pins.



Screen 2: Space Details (18×9.5 ft, EV Charger, CCTV) & Instant UPI Booking Calculator.



Screen 3: Driver Trips Hub & Digital Parking Pass with 4-digit Gate PIN (4829).



Screen 4: Host Command Center displaying earnings chart, active bays, and net payouts.

The prototype consists of four core functional views:

1. **Live Geospatial Discovery Map:** Enables drivers to filter by Pune neighbourhoods, vehicle size, and amenities while comparing hourly rates directly on an interactive Leaflet map.
2. **Space Specifications & Booking Calculator:** Displays verified host credentials, bay clearance dimensions, and real-time hourly fee breakdowns (base rent + 5% service fee) with slot collision prevention.
3. **Driver Trips & Digital Gate PIN Pass:** Presents active reservations alongside a dark-themed pass showing the 4-digit Gate PIN (4 8 2 9), booking reference, and gate entry instructions.
4. **Host Workspace & Admin Governance:** Provides hosts with a pin-drop space onboarding wizard and 30-day revenue analytics, paired with an Admin Trust & Safety view for resolving reported access issues.

The prototype communicates the application's layout, navigation, and core features. It demonstrates how the solution makes private parking availability more transparent, supports better commute planning, and encourages community participation in sharing underutilized urban infrastructure.

Note: These screens are captured from the working ParkOrbit web prototype using seeded Pune geospatial data and an embedded multi-persona simulator. Field testing with real residential societies and live UPI payment gateways would be the next step to evaluate and improve the solution.

Conclusion

The Design Thinking process helped us systematically explore a practical urban mobility problem faced by daily commuters and property owners in Pune's high-density corridors. Through the Empathize, Define, Ideate, and Prototype phases, we developed **ParkOrbit — Peer-to-Peer Urban Parking Marketplace**.

The application helps users check available private parking spaces and EV chargers on an interactive map, verify vehicle clearance and pricing upfront, check in securely using a 4-digit Gate PIN, and report access barriers. At the same time, it enables homeowners to list empty driveways effortlessly and earn passive income.

This activity strengthened our understanding of user-centred problem-solving, structured idea evaluation, and full-stack prototype design. Overall, the project demonstrates how design thinking and geospatial technology can turn dormant private parking bays into accessible, sustainable, and community-driven urban mobility infrastructure.